



Experimental realization of an all-optical computing system using a coherent nonlinear scattering medium

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ABSTRACT

Artificial neural networks are indispensable tools across various fields, but their increasing complexity and energy demands have motivated the development of optical neural networks (ONNs) for faster and more efficient computation. In this work, we present an all-optical nonlinear ONN for image classification, employing a second-harmonic generation scattering medium and two spatial light modulators (SLMs). The first SLM encodes input images, and the SHG medium performs nonlinear mapping, while the second SLM functions as a trainable classification layer. Validated on the MNIST dataset, our system demonstrates the feasibility of all-optical inference, highlighting its potential for energy-efficient and high-speed computing.

1. Introduction

Artificial neural networks (ANNs) are an essential tool for diverse fields, including image analysis [1], autonomous driving [2], language processing [3], and medical diagnostics [4]. However, as these computational devices become increasingly larger and more complex, their high costs and energy consumption have prompted the search for more efficient methods to construct neural networks. This has culminated in the development of optical neural networks (ONNs), promising several advantages like extremely fast computation speed (low latency), efficient energy consumption, and high parallelism [5,6]. These advantages are leveraged by using optical components for computing, such as digital micromirror devices (DMD) [7], SLM spatial light modulators (SLMs) [8] and other diffractive elements [9–13], along with the light guiding media like multimode fibers [14] and nonlinear crystals [15].

Despite its distinct advantages, implementing all-optical ONNs faces several challenges, particularly in achieving nonlinearity and tunability [11,13]. Compared to electronic networks, optical elements are usually not inherently tunable. To address this, extreme learning machines (ELMs) have emerged as a viable alternative to traditional neural network architectures. This approach utilizes fixed mappings through the layers, with only the readout layer being trainable [16,17]. Therefore, a typical ONN based on ELM implements the following operations: input encoding, linear mapping, nonlinear activation, and classification using a trainable readout layer [8]. To introduce optical nonlinearity, several methods have been reported [18], such as the use of atomic

interactions [19], recurrent scattering [20], image intensifiers [21], and nonlinear optical effects [22]. However, in most studies, one or more of the key ONN operations are performed on a computer rather than being fully implemented optically. For example, in a recent study, optical nonlinearity was implemented using second harmonic generation (SHG), but the trainable readout layer was still implemented digitally [23]. To fully harness the advantages of ONNs, it is crucial to achieve complete optical implementation of all operations, including nonlinearity and readout, while minimizing dependence on computational systems [24].

In this work, we develop and implement an all-optical nonlinear neural network for image classification. The core of our ONN comprises an SHG scattering medium and two spatial light modulators (SLMs). The first SLM encodes the input image, and the scattering medium performs nonlinear mapping to the phase-encoded input image through the coherent SHG process [25]. The second SLM acts as a trainable classification (readout) layer, with the output captured by a camera. The network is trained to focus input images with specific labels on designated regions of the camera.

The proposed nonlinear ONN's image classification capabilities were validated using the MNIST dataset. We first experimentally generated a training dataset and computationally derived the optimal readout layer. Once the optimized layer was implemented on the second SLM, the nonlinear ONN performed all-optical inference without relying on digital computation. Although the accuracy of optical inference was lower than that of digital inference, the system demonstrated the ability to

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predict many labels with high fidelity, confirming the feasibility of realizing an all-optical nonlinear ONN. Notably, during network training, we recorded the SHG field using interferometric detection and proposed the coherent field sum as an optimization metric, improving the training fidelity compared to the typical incoherent sum. This approach also underscores the potential for using the ONN's output to enable additional optical operations, further emphasizing the versatility and significance of our implementation.

2. Results

2.1. Experimental setup of the optical computing system and its operation procedures

Our optical computing system is designed for image classification. The schematic diagram of the experimental setup is shown in Fig. 1. First, we sent intense pulses from a regenerative amplifier laser (Coherent Libra; center wavelength, $\lambda_F = 800$ nm; pulse width, 91 fs; bandwidth, 15 nm; repetition rate, 10 kHz) to induce SHG signals over a wide field of view. The output beam from the laser passes through a spatial light modulator (SLM1, Hamamatsu X13138-01), which encodes the input image x as a phase map. Note that the image is normalized between 0 and 1.8π , and scaled to 50 by 50 pixels before being displayed on SLM1. The encoded field, represented as $E_{in} = \exp(ix)$, is projected to a scattering medium composed of randomly dispersed BaTiO₃ particles.

This medium performs nonlinear mixing of the 2500 incident channels, which produces an SHG wave E_{SHG} with 81×81 output channels, corresponding to the number of resolvable optical modes ($\lambda_{SHG} = 400$ nm) [25]. E_{SHG} is then sent through another spatial light modulator (SLM2) which adds the phase retardation β to the output channels, i.e. $e^{i\beta} \cdot E_{SHG}$. This field is then Fourier transformed by a lens, and the final output field E_{out} is retrieved using digital holography. The arrangement of this experimental setup is equivalent to a computing network described in Fig. 1b. In our network, the trainable parameter is β , in accordance with the framework of an ELM. However, note that although inspired by the ELM framework, our approach trains the classification layer using gradient-based iterative optimization.

The actual implementation of our optical computing system is as follows. The input image is upscaled to 50×50 pixels and encoded as a phase map on SLM1 (400×400 pixels, $12.5 \mu\text{m}$ pixel pitch), such that each encoded pixel covers an 8×8 region on SLM1. The phase-encoded field is projected onto the nonlinear scattering medium, with the field of view being $45 \times 45 \mu\text{m}^2$, corresponding to 50×50 input channels. The generated SHG wave is sampled over a field of view of $40 \times 40 \mu\text{m}^2$, corresponding to 81×81 output channels, determined by the system's numerical aperture ($NA = 0.4$) at 400 nm. The SHG field is then relayed onto SLM2, which operates at a physical resolution of 173×173 pixels ($12.5 \mu\text{m}$ pixel pitch). The classification-layer phase map (β), trained at 81×81 pixels, is interpolated to match the SLM2 resolution for phase modulation of the SHG field. The Fourier transform of the modulated

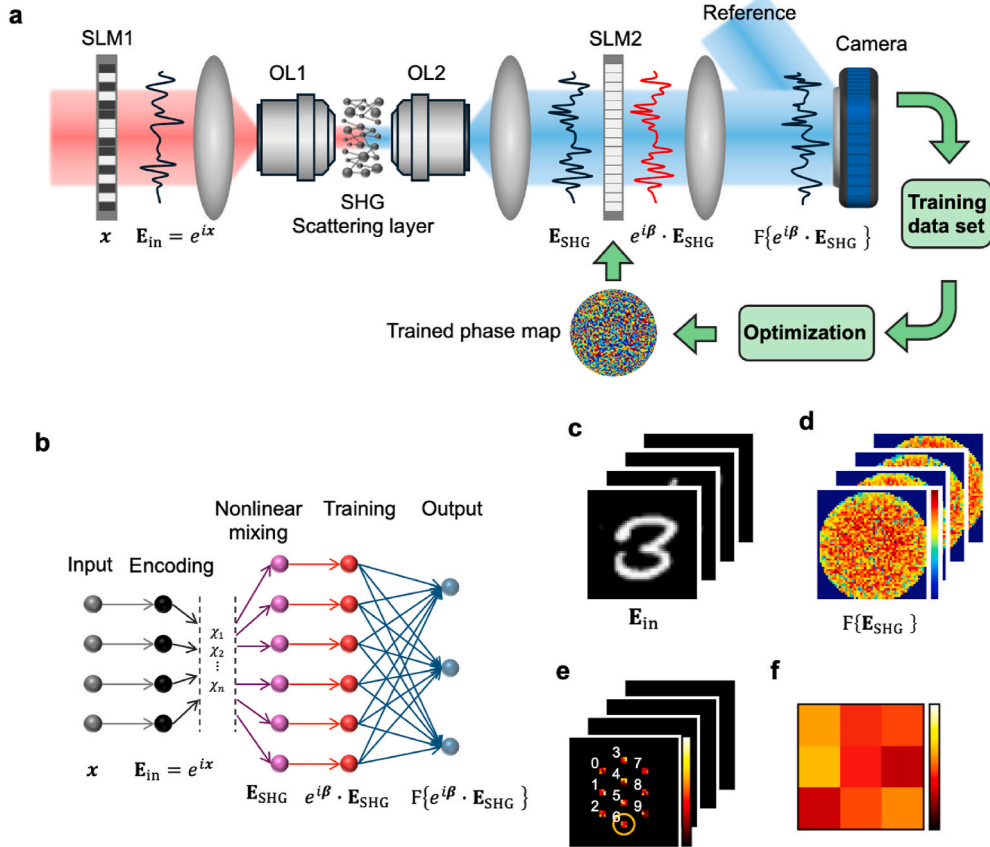


Fig. 1. Schematic of the experimental setup and training procedures. (a) Schematic of the experimental setup. SLM1 and 2: phase-only spatial light modulators. OL1 and 2: objective lenses. The SHG scattering layer is made of randomly dispersed BaTiO₃ particles. A reference wave is introduced to record complex field maps. SLM1 displays the phase map, x , a vectorized grayscale image. E_{in} and E_{SHG} : phase-encoded input data and output SHG wave, respectively. Training data is acquired while using SLM2 as a flat mirror. After the training, the trained phase map β is displayed on SLM2 for inference. $F\{\}$: Fourier transform operation by a lens. (b) Optical computing model implemented by the experimental setup in (a). The model comprises an input layer, an encoding layer, a nonlinear mixing layer, a training layer, and an output layer. (c) Phase-encoded data set written on the SLM1. (d) SHG speckle maps captured by the camera. Only the amplitude maps are shown while there were associated phase maps. (e) Positions of individual numbers used for training from the images in (d). (f) Zoom-in image of the area indicated by a circle in (e). Each position consists of 9 pixels.

field is captured by the camera, with digital holography used to reconstruct the complex optical field.

The operation of our ONN consists of three steps – experimental acquisition of the training data set, computational training to find β , and the optical realization of the trained network by displaying β on the SLM2. For the first step, the system records a training dataset using a set of MNIST images sequentially displayed on SLM1 (Fig. 1c). The E_{out} was recorded for 1100 MNIST images, which were equally distributed between each label. During the training phase, the phase map on SLM2 is set to $\beta = 0$, and a reference beam with a wavelength of 400 nm is introduced to the camera for recording an interferogram and retrieving the complex field map of E_{out} , effectively giving us E_{SHG} . The reference beam is generated via SHG using a 0.5 mm-thick BBO crystal [25]. Fig. 1d shows the recorded training dataset. For each recorded map, we selected 10 subregions corresponding to the classification labels ranging from 0 to 9 (Fig. 1e). Each subregion consists of 9 pixels, as shown in Fig. 1f. We defined the readout value for a given label i as the squared magnitude of the coherent sum of the complex field values over the 9 pixels in the corresponding subregion, i.e., $I_{\text{pred}}(i) = \left| \sum_{j=1}^9 E_{\text{out}}(i_j) \right|^2$, where i_j denotes the individual pixels of the i th subregion. This is equivalent to placing a microlens array for each subregion and measuring the intensity at the focal plane of the microlenses. The label corresponding to the subregion with the highest readout value is the prediction readout value.

For the second step, referred to as computational training, we aim to find the optimal β for SLM2, which minimizes the difference between the actual labels and the predicted readout values for each handwritten image. With the training dataset, β was obtained using optimization on a personal computer (CPU: Intel i7-7700k, RAM: 64 GB), utilizing the ADAM optimizer in PyTorch (500 epochs, 10 batches, learning rate 0.1; 0.3 s per epoch). The Pearson correlation between the predicted readout values and the one-hot encoded ground-truth labels was used as a loss function. For each label, 1000 images were used for training, and 100 images were used for validation. For the third step, which is optical computing, the optimal trained β was physically displayed on SLM2 to measure E_{out} , and the ensuing predicted labels were compared with the actual labels corresponding to different input images.

2.2. Choice of optimization metric in the training process

In many previous studies, the incoherent sum of intensities for each pixel in a subregion was used as the predicted label for the optimization metric [9], i.e., $I_{\text{pred}}(i) = \sum_{j=1}^9 |E_{\text{out}}(i_j)|^2$. This approach provides higher fidelity compared to using the intensity value of a single pixel, due to averaging random noise over adjacent pixels. However, in our study, we

used the squared magnitude of the coherent field sum as the predicted label [12]. The coherent sum leads to an enhanced signal-to-noise ratio (SNR) in proportion to the number of pixels. Its SNR is higher than that of the incoherent sum, which scales with the square root of the pixel number. This benefit is validated in Fig. 2, which compares the training results for both cases. Fig. 2a and b presents the total energy distribution over the predicted labels when the incoherent intensity summation and coherent field summation are used as the optimization metric, respectively. Note that the optimized β is different for the two metrics. For ease of analysis, the total energy is normalized to 100 for each result, i.e., $\sum_{i=0}^9 I_{\text{pred}}(i) = 100$. In both cases, the diagonal terms—where the predicted label matches the true label—have the highest values, corroborating the efficacy of both choices. However, in the case of the coherent field summation, the average value is 56.9, more than twice the 26.5 achieved with the incoherent intensity summation.

Fig. 2c shows the energy distribution across the predicted labels for a representative example where the true label is 1. The SNR for the coherent summation is 3.53 times higher than that of the incoherent summation. For the entire training dataset, the SNR for the coherent summation was 11.88, which is 3.66 times higher than the 3.25 observed for the case of the incoherent summation. This higher SNR is especially critical in practical experimental realizations, where the presence of experimental noise makes configurations with higher SNRs more feasible to implement. Thus, we chose the coherent field sum as the metric for our experiments.

2.3. Optical computing outcomes

Using the coherent sum as the optimization metric, we evaluated the performance of both computational training and optical computing. We employed Pearson loss for training, as it was found to outperform cross-entropy loss in our application. Fig. 3a and b shows the confusion matrices for the training and validation datasets after the computational training to obtain the optimized β . The training dataset achieved 100 % accuracy, while the validation dataset showed 80 % accuracy. The evolution of loss over epochs is shown in Fig. 3c. Here, accuracy is defined as the ratio of correct classifications to the total number of images. Note that in this case, β was digitally applied to the experimentally measured E_{SHG} to compute the predicted readout values for classification. Fig. 3d provides a representative example, displaying the energy for each label for a handwritten image labeled as “1” in the training dataset. The energy values are normalized with respect to the peak for clear visualization. The label was correctly predicted with 96.1 % energy focused on the true label.

Fig. 3e and f presents the confusion matrices obtained from optical computing for the training and validation datasets, respectively, after

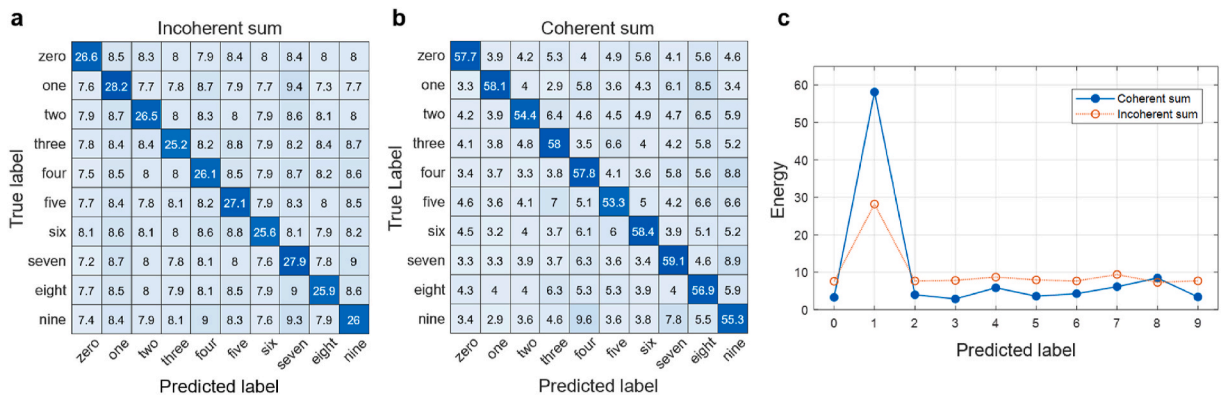


Fig. 2. Training fidelity depending on the incoherent intensity and coherent field sums as the training metric. (a–b) The energy distributions over the predicted labels for incoherent intensity sum and the absolute square of the coherent field sum. Results were obtained from 1000 handwriting images with the energy distributions normalized to 100. (c) Energy distribution for both methods for representative data with a true label of 1. The blue solid dots represent simulation results using field sum, while the orange open circles represent simulation results using intensity sum.

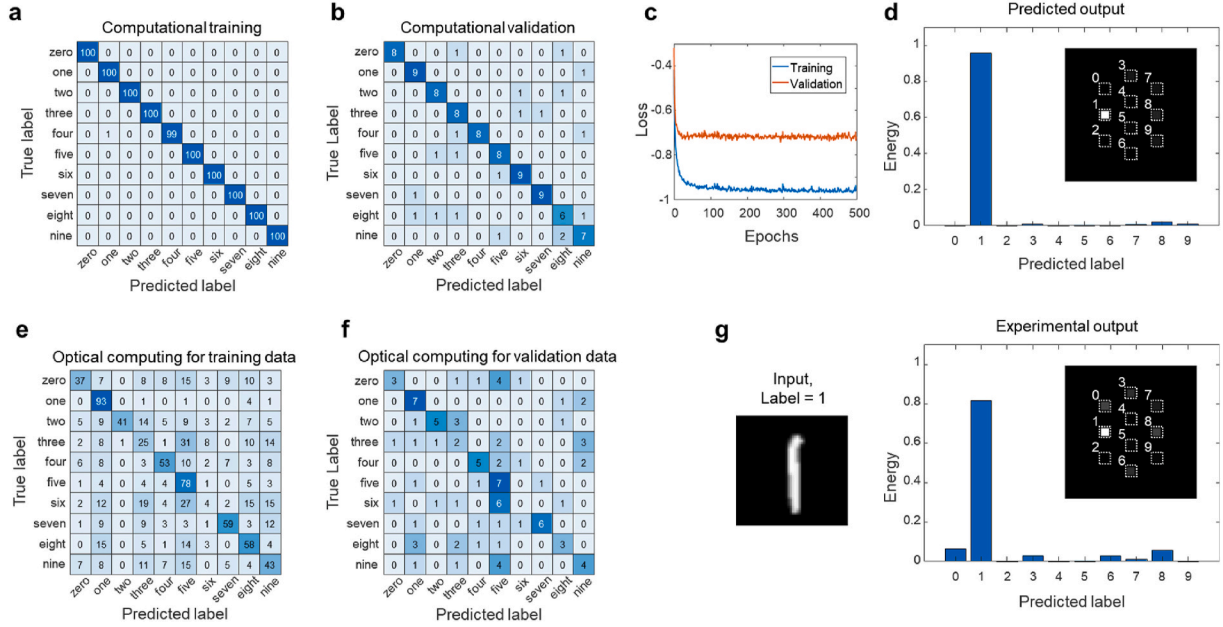


Fig. 3. Optical computing results. (a–b) Confusion matrices of computational training and validation. (c) Variation of loss over epochs. (d) Example of predicted output from computational training corresponding to an input image with label '1'. (e–f) Confusion matrices of optical computing for the same training and validation sets in (a) and (b), respectively. (g) The output of the optical computing for the same input image as (d). The input image is also shown for reference.

physically displaying the trained β on SLM2. From the recorded E_{out} , the accuracies for these cases were found to be 49.1 % and 42.0 %, respectively, both lower than the results of computational training. Fig. 3g illustrates the predicted output for the same input image as in Fig. 3d, obtained through optical computing. In this case, the energy focused on the correct label was 81.6 %, less than obtained during computational training. The lower fidelity of optical computing compared to computational training is primarily attributed to the imperfect realization of phase modulation by SLM2.

2.4. Error analysis

Although our optical computing system successfully implemented MNIST recognition optically, its fidelity fell short compared to computational training. The primary reason for this limitation is the experimental inaccuracy of SLM2 in implementing the trained β . Regardless of the prediction accuracy, the Pearson correlation between the output field $e^{i\beta} \cdot E_{\text{SHG}}$ obtained computationally using the measured E_{SHG} and trained β , and the experimentally measured output field after displaying β on SLM2, was approximately 50 %. This suggests that the implementation accuracy of SLM2 was around 50 % on average for the entire dataset.

The spatial light modulator (SLM) used in this study (Hamamatsu X13138-01) operates within a wavelength range of 400–700 nm. Consequently, the SHG wavelength used in SLM2 was near the edge of its operational range. As a result, although the SLM theoretically supports 256 levels of phase modulation (0–255), approximately 87 discrete phase levels were available at the wavelength of 400 nm. Additionally, while the trained β was of size 81×81 pixels, the SLM physically displayed the phase at a resolution of 173×173 pixels, requiring interpolated mapping to facilitate the experimental implementation of SLM2. However, it was verified that this did not negatively impact the performance. The reduction of accuracy may be attributed to a potential lateral misalignment of the SLM2. For instance, during training, it was found that even a single pixel misalignment resulted in a decrease in the accuracy to 36 %. Although efforts were made to minimize such misalignment, a residual error of this magnitude in the optical system could have contributed to the reduction of performance of optical

computing.

Interestingly, the optical computing results varied widely from image to image, as shown in Fig. 4. In some cases, the experimental results closely matched with that of the computational training (Fig. 4a). In some cases, they were marginally consistent, which means that the correct label was predicted even though the computational and optically obtained energy distributions across the labels did not match well (Fig. 4b). In other cases, even the predicted label was incorrect (Fig. 4c). However, in all three cases, the experimental implementation of $\exp(i\beta)$ had similar precision, as seen from the phase maps of $F^{-1}\{E_{\text{out}}\}$ presented in the insets in Fig. 4a–c. We chose to compare $F^{-1}\{E_{\text{out}}\}$, rather than E_{out} , because it corresponds to the field at a plane conjugate to the SLM. The computationally optimized predictions (left), optically obtained predictions (middle), and their difference (right) are shown. We expect that with a more accurate mapping of the optimized classification layer to the readout SLM, the experimentally achieved accuracy would approach the computationally obtained one.

3. Discussion

In this study, we demonstrated an all-optical nonlinear neural network (ONN) for image classification using a coherent nonlinear scattering medium. By employing a second-harmonic generation (SHG) scattering medium and spatial light modulators (SLMs), we implemented optical operations for encoding, nonlinear mapping, and classification without relying on digital computation. This approach highlights the potential of ONNs to reduce energy consumption and computation latency by leveraging the inherent advantages of optical systems, such as high parallelism and speed.

Our results showed that computational training of the classification layer achieved high accuracy on the MNIST dataset, validating the design of our system. Although the optical implementation exhibited lower accuracy, it still demonstrated the feasibility of fully optical inference, with many predictions retaining high fidelity. The reduced performance in optical inference was primarily due to the implementation of β using SLM2. The main cause was attributed to potential misalignment of SLM2, despite careful efforts to ensure alignment and precision. This led to an average implementation accuracy of approxi-

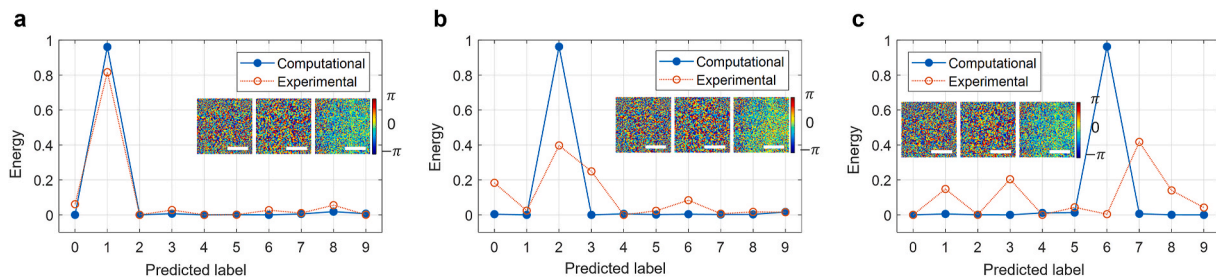


Fig. 4. Comparison between computational and optical computing outputs. (a) A representative case where the computational prediction is successfully implemented in optical computing as well. (b) Same as (a), but for a case where the optical results are marginally consistent with the computational predictions. Although the energy distributions do not exactly match, the label is predicted correctly. (c) In this case both the energy distribution and the predicted labels do not match. Insets show the optically (left) and computationally (middle) obtained $F^{-1}\{E_{out}\}$ as well as their difference (right). Notably, the correlation between the images is similar for all three cases. Scale bar: 15 μm .

mately 50 %, which affected the overall fidelity of the system. There were also inherent challenges such as the limitation of the SLM in achieving precise phase modulation at the operational wavelength, and the necessity of interpolation during phase mapping. Although these factors did not significantly affect the results, they remain important considerations in the system's overall performance. The accuracy can also be improved by training β *in situ* rather than using the current *in silico* approach, which would allow for direct physical tuning of SLM2 based on the optical output during training [11,15,24].

A key feature of our work was the use of the coherent field sum as the optimization metric during training, which improved the training fidelity by more than three times compared to traditional incoherent intensity summation methods. The enhanced fidelity was crucial in mitigating noise in experimental environments, enabling more robust optical operations and higher classification accuracy. Since the training and inference were based on the SHG field rather than the intensity, the ONN's output can be utilized for additional optical operations, thereby maximizing the advantages of optical computing. It should be noted that although holography was used to facilitate training, it is not required for inference—by positioning a microlens array to focus each subregion onto a single bucket detector, we can achieve the coherent field sum without relying on holography. Future efforts to improve phase modulation accuracy, such as employing SLMs with higher resolution or broader operational wavelength ranges, could significantly enhance performance. The ability of our ONN to achieve nonlinear mapping and classification entirely optically paves the way for integrating additional optical operations, further expanding its applications in high-speed and energy-efficient computing.

CRediT authorship contribution statement

Ye-Chan Cho: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Kalpaka Gupta:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation. **Jungho Moon:** Resources, Methodology, Conceptualization. **Wonshik Choi:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

Code availability

The code used in this study is available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available from the authors upon reasonable request.

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